

Digital Signal Processing

Introduction to Discrete-time Signals and Systems

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Signals, Systems and Signal Processing

- A signal can be defined as a physical quantity that varies with time, space or any independent variable

Case 1

$$s_1(t) = 5t$$

Signal varies linearly with one independent variable 't'

Case 2

$$s_2(t) = 20t^2$$

Signal varies quadratically with one independent variable 't'

Case 3

$$s(x,y) = 3x + 2xy + 10y^2$$

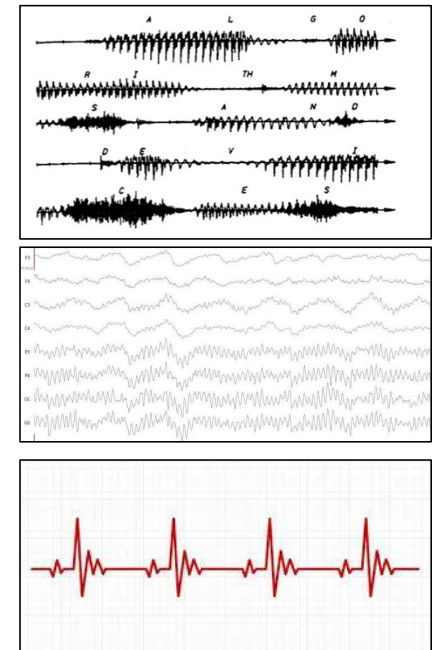
Signal varies according to two independent variables x and y

- In real life, signals mostly have a highly complicated functional relationship with independent variables. For eg. Speech signals
- These kinds of signals are represented as a sum of several sinusoids of different amplitudes and frequencies.

$$\sum_{i=1}^N A_i(t) \sin[2\pi F_i(t)t + \theta_i(t)]$$

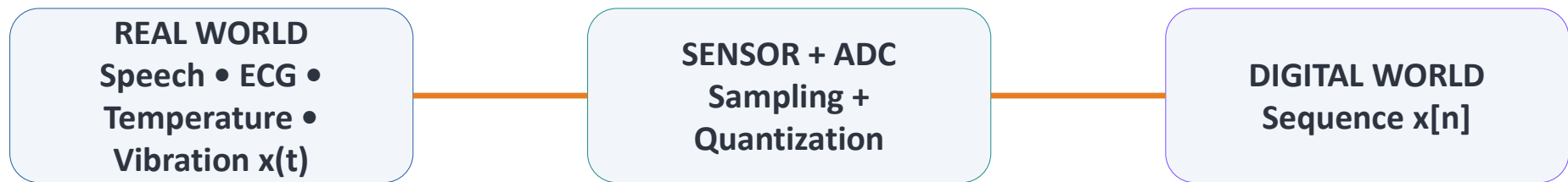
where, $\{A_i(t)\}$, $\{F_i(t)\}$, and $\{\theta_i(t)\}$ are sets of amplitudes, frequencies and phases of the sinusoids

- Other examples is ECG and EEG that evolve as a function of a single variable 't'
- An example of a two-variable signal??



Signals, Systems and Signal Processing

- Signal generation is associated with a system that responds to a stimulus or force
- A system is a physical device that performs an operation on a signal. For eg., a filter used to reduce the noise
- When a signal is passed through a system, such as in filtering, we say that we have processed the signal.



- Computers and digital processors do not directly manipulate an analog waveform $x(t)$.
- They operate on a list of numbers obtained at specific time instants.. That ordered list is a discrete-time sequence.

Continuous Time vs Discrete Time signals

Continuous-time $x(t)$

Defined for every real-valued time t

Example: temperature changing continuously

- Defined for **every instant of time**.
- Represented as $x(t)$, where t is a continuous variable.
- Usually obtained from physical phenomena such as sound, temperature, and voltage
- Example: $x(t) = A \sin(2\pi ft)$

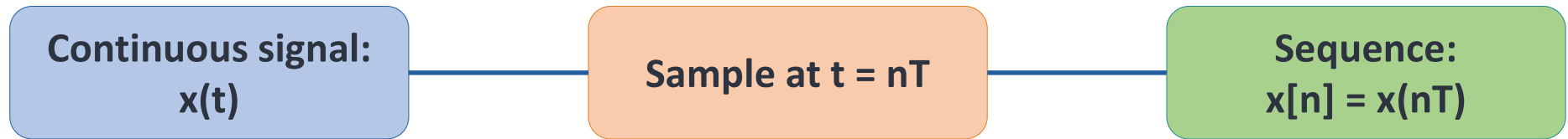
Discrete-time $x[n]$

Defined only at integer indices n

Example: temperature measured every minute

- Defined only at **specific discrete time instants**
- Represented as $x[n]$, where n is an integer
- Usually obtained by sampling a continuous-time signal.
- Example: $x[n] = A \sin(2\pi fn)$

From a Continuous Signal to a Sequence



Sampling converts the time axis from continuous to discrete.

The amplitude values are still continuous at this stage.

Quantization is a separate step that makes amplitudes discrete too.

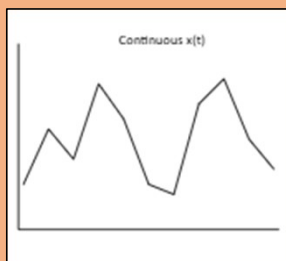
Analog to Digital

Step 1: Real world Analog signal

Physical signals usually vary continuously with time.

Examples: microphone voltage, ECG, temperature sensor output.

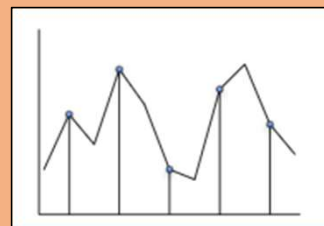
We represent an analog signal as $x(t)$, where t can take any value.



Step 2: Sampling (Measure at specific instants)

Sampling means taking the value of $x(t)$ at regular time intervals. If the sampling interval is T_s , samples occur at:
 $t = 0, T_s, 2T_s, 3T_s, \dots$

The sampled sequence is written as: $x[n] = x(nT_s)$



How often do we take samples?

Sampling frequency:

$$f_s = 1/T_s \text{ samples/second.}$$

Higher f_s means more Samples and a better representation of fast changes.

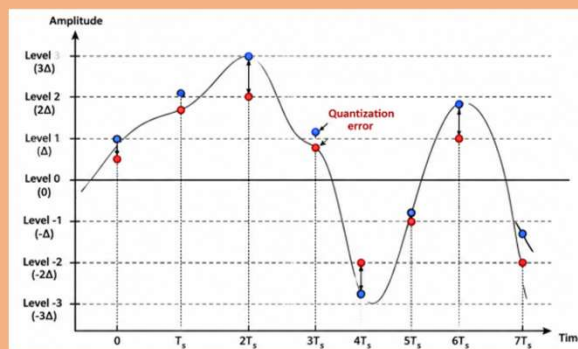
Too low a sampling rate can lose information and create **aliasing**.

Nyquist principle: for a band-limited signal, f_s should be at least **twice the highest frequency component**.

Step 3: Quantization (Convert Each Sample to a Fixed Level)

After sampling, a sample can still have many possible analog values.

Quantization rounds each sample to the nearest allowed amplitude level. The result is an approximate value chosen from a **finite set of levels**. The difference between the original sample and the quantized value is **quantization error**.



Step 4: Encoding — Represent the Quantized Value in Binary

Each quantized level is assigned a **binary code**. An **N-bit ADC** can represent 2^N different levels. **Example:**

3-bit ADC $\rightarrow 2^3 = 8$ levels

The final output is now a **stream of bits** that a computer can store and process.

Key idea:

Quantization $\rightarrow$ Choose a level

Encoding $\rightarrow$ Give that level a binary number

Quantization Level	Level Index	3-bit Binary Code
Level 3	0	000
Level 2	1	001
Level 1	2	010
Level 0	3	011
Level -1	4	100
Level -2	5	101
Level -3	6	110
	7	111

Discrete-Time Signal $\neq$ Digital Signal

Discrete-time signal
Time/index is discrete
Amplitude may be continuous

Sampling $\rightarrow$ makes time
discrete

Digital signal
Time/index is discrete
Amplitude is also quantized

Quantization $\rightarrow$ makes amplitude
discrete

1. Sensor
Physical quantity $\rightarrow$
voltage

2. Sampling
Measure at discrete
times

3. Quantization
Round to nearest level

4. Encoding
Convert level to bits

5. Digital Signal
Sequence $x[n]$

Formal Definition of a Discrete-Time Signal

A discrete-time signal is a function of an integer-valued independent variable:
 $x[n], n \in \mathbb{Z}$

n is an index, not necessarily a physical time variable.

If samples are taken every T seconds, the n th sample corresponds to time $t = nT$

The values $x[n]$ may be real or complex.

A finite sequence: $x[n] = \{x[0], x[1], \dots, x[N-1]\}$.

Discrete time sinusoidal signals

- A discrete-time sinusoidal signal may be expressed as

$$x(n) = A \cos(\omega n + \theta) \quad \text{where } -\infty < n < \infty$$

where n is an integer variable, called sample number, A is the amplitude of the sinusoid, ω is the frequency in radians per sample and θ is the phase in radian.

- Also, $\omega = 2\pi f$
- Therefore, $x(n) = A \cos(2\pi f n + \theta)$ where $-\infty < n < \infty$
- A discrete time sinusoid is periodic only if $x(n + N) = x(n)$ for all n
- The smallest value of N for which the above is true is called the fundamental period. It is the *smallest* positive value of N (or T for continuous-time) for which a signal exactly repeats itself.

Consider the Sinusoid:

$$x[n] = 2 \cos(0.2\pi n + 0.9\pi)$$

It is periodic with period $N=10$ since

$$\begin{aligned} x[n+10] &= 2 \cos(0.2\pi(n+10) + 0.9\pi) \\ &= 2 \cos(0.2\pi n + 0.9\pi + 2\pi) = x[n] \quad \text{for all } n. \end{aligned}$$

- Cosine has a fundamental period of 2π , adding any integer multiple of 2π to its argument leaves the value unchanged.
- Here, shifting n by 10 samples adds exactly 2π times 10 = 2π to the argument — a full cycle — so the signal repeats.

How Do We Represent a Sequence?

Tabular form

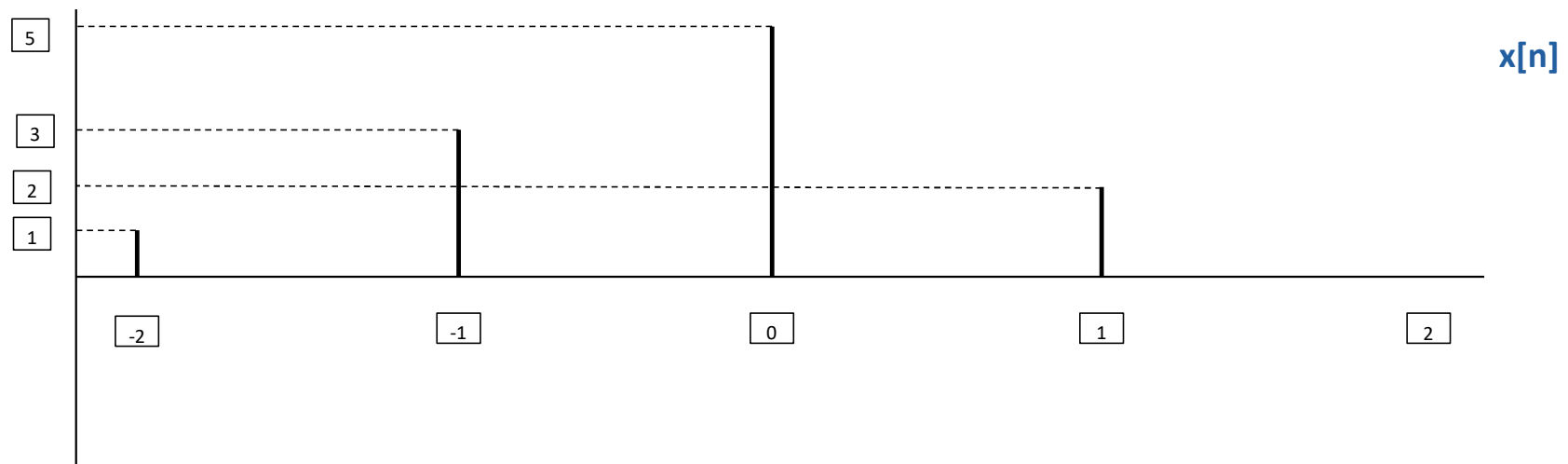
n: -2 -1 0 1 2
x[n]: 1 3 5 2 0

Set notation

$x[n] = \{1, 3, \uparrow 5, 2, 0\}$
 $\uparrow$ marks $n = 0$

Graphical form

Stem plot: each sample is represented by a vertical line and a marker.



A discrete-time sequence is not a continuous curve

Standard Sequences: The Building Blocks

Unit impulse sequence $\delta[n]$

Unit step sequence $u[n]$

Ramp sequence $r[n]$

Real exponential $a^n u[n]$

Complex exponential
 $Ae^{j\omega_0 n}$

Sinusoidal sequence $A \cos(\omega_0 n + \phi)$

1. Unit Impulse Sequence $\delta[n]$

- Also called the Kronecker delta.
- Represents a single sample located at the origin.
- Shifting gives $\delta[n - n_0]$, an impulse located at $n = n_0$.

- The simplest interpretation of the Kronecker delta is the discrete version of the delta function defined by

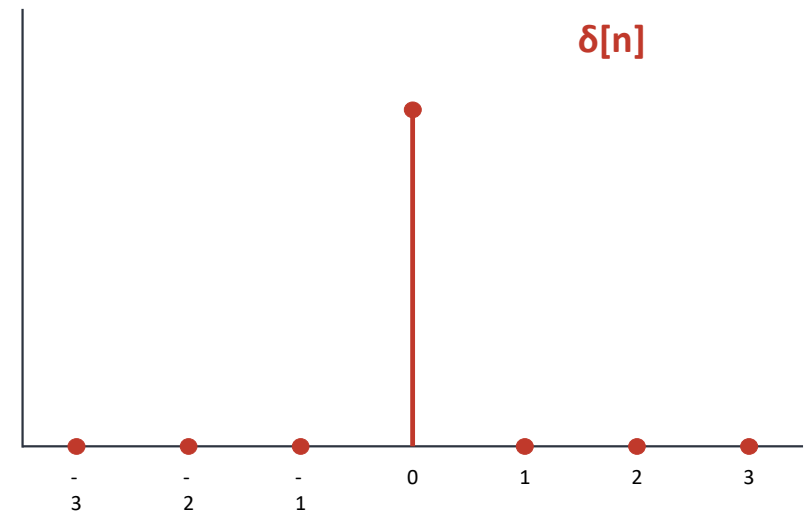
$$\delta_{ij} \equiv \begin{cases} 0 & \text{for } i \neq j \\ 1 & \text{for } i = j. \end{cases}$$

- **The unit impulse is the basic building block of all discrete-time signals.**

For example, if $x[n] = [2, 4, 3]$

Then

$$x[n] = 2\delta[n] + 4\delta[n-1] + 3\delta[n-2]$$



$$\begin{aligned} \delta[n] &= 1, \quad n = 0 \\ \delta[n] &= 0, \quad n \neq 0 \end{aligned}$$

2. Shifted Impulse: $\delta[n - k]$

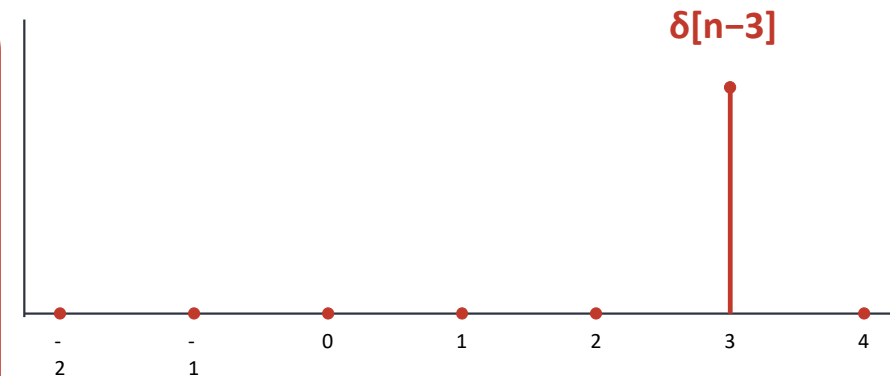
Shifting

If the signal $\delta[n]$ is shifted by k samples, you get $\delta[n - k]$, which is 1 only when $n = k$:

$$\delta[n - k] = \begin{cases} 1, & n = k \\ 0, & n \neq k \end{cases}$$

So:

- $\delta[n] \rightarrow$ spike at $n = 0$
- $\delta[n - 1] \rightarrow$ spike at $n = 1$
- $\delta[n - 2] \rightarrow$ spike at $n = 2$



$\delta[n - 3]$ is nonzero only at $n = 3$

Scaling

If $\delta[n - k]$ is multiplied by some constant a , you get a spike of height a located at $n = k$:

$$a \delta[n - k] = \begin{cases} a, & n = k \\ 0, & n \neq k \end{cases}$$

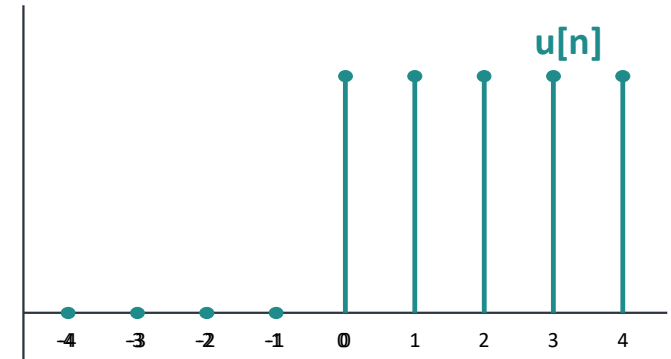
Any discrete-time sequence can be written as a sum of scaled, shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

3. Unit Step Sequence $u[n]$

A sequence that 'turns on' at $n = 0$

$$u[n] = 1, \quad n \geq 0$$
$$u[n] = 0, \quad n < 0$$



$u[n-k] \rightarrow$ Shift right by k

$u[n+k] \rightarrow$ Shift left by k

Imagine an automatic machine:

$u[n]$ = Machine turns ON at $n=0$

$u[n-3]$ = Machine turns ON after 3 time units

Relationship Between Impulse and Step

$$u[n] = u[n-1] + \delta[n]$$

$$\delta[n] = u[n] - u[n-1]$$

The impulse is the first difference of the step.

The step can be interpreted as the accumulated sum of impulses.

This relationship becomes important in difference equations and discrete-time systems.

4. Discrete-Time Ramp Sequence

Starts at zero.

Increases linearly for positive n .

Used to model cumulative growth.

$$r[n] = n u[n]$$

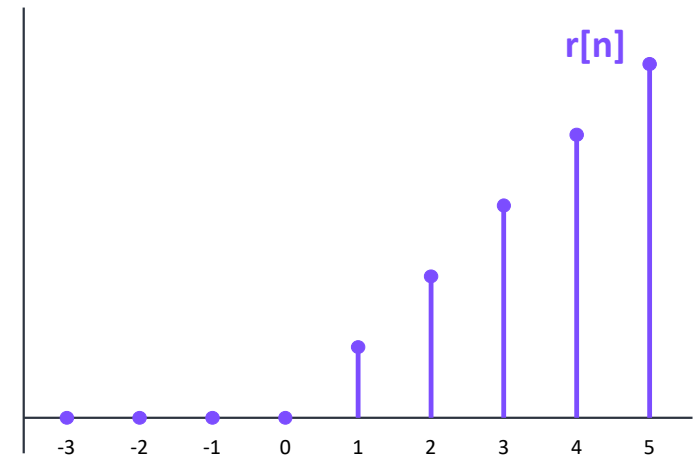
For $n < 0$, $u[n]=0$, so $r[n]=0$

For $n \geq 0$, $u[n]=1$, so $r[n]=n$

Shifted Ramp

$$r[n-2] = (n-2)u[n-2]$$

This ramp **starts increasing at $n=2$** instead of $n=0$.



$$r[n] = n u[n]$$

6. Discrete-Time Sinusoidal Sequence

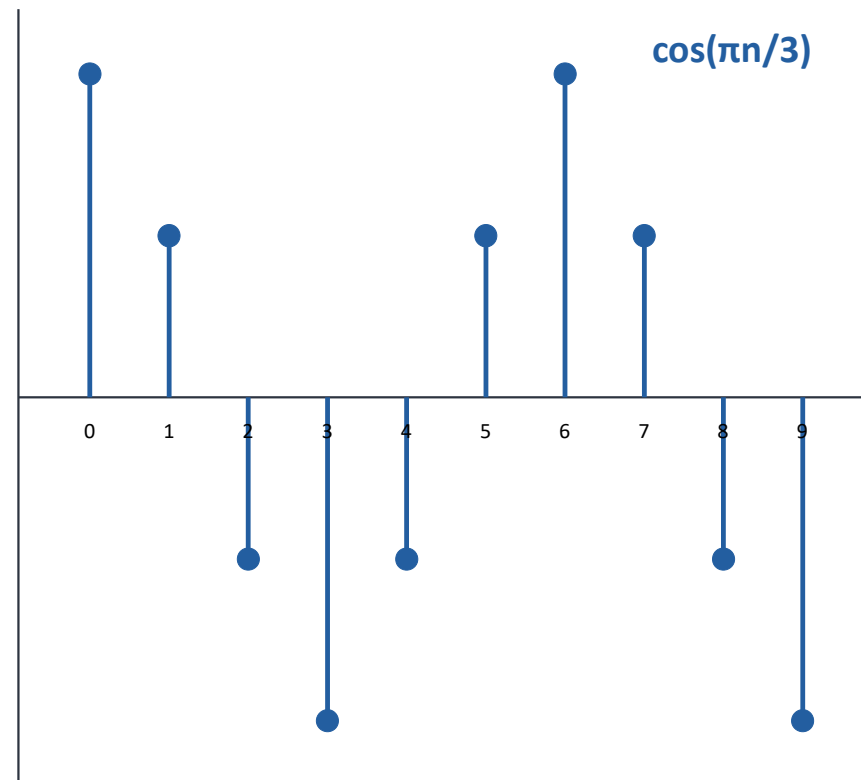
$$x[n] = A \cos(\omega_0 n + \phi)$$

A controls amplitude.

ω_0 is digital frequency in radians/sample.

ϕ is phase.

Unlike continuous time, periodicity depends on whether $\omega_0/(2\pi)$ is rational.



7. Complex Exponential Sequence

$$x[n] = A e^{j\omega_0 n}$$

Euler identity:

$$e^{j\omega_0 n} = \cos(\omega_0 n) + j \sin(\omega_0 n)$$

Fundamental building block for Fourier analysis.

Its magnitude is constant when A is constant.

The angle rotates by ω_0 radians for every increment of n .

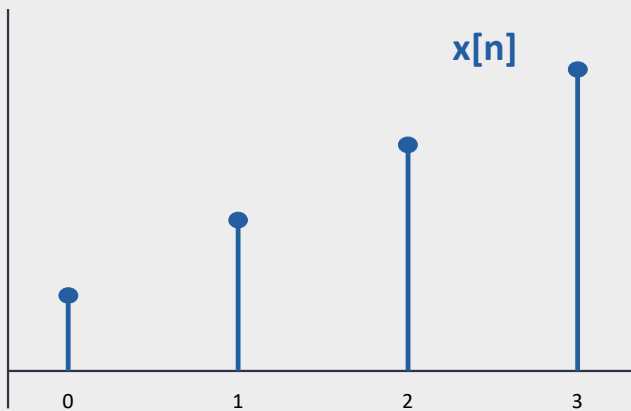
Discussion

1. Draw $\delta[n-2]$

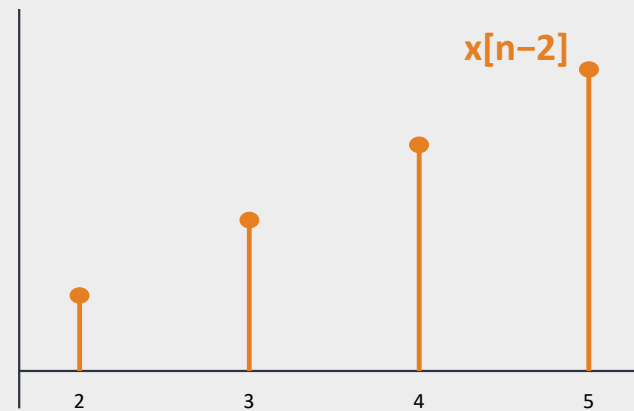
2. Draw $u[n+3]$

3. For $x[n]=2(0.5)^n u[n]$, list $n=0\ldots 4$

4. Given $x[n] = \{\uparrow 1, 2, 3, 4\}$



Find $y[n] = x[n-2]$



Every sample moves 2 units to the right.

Time Reversal (Folding)

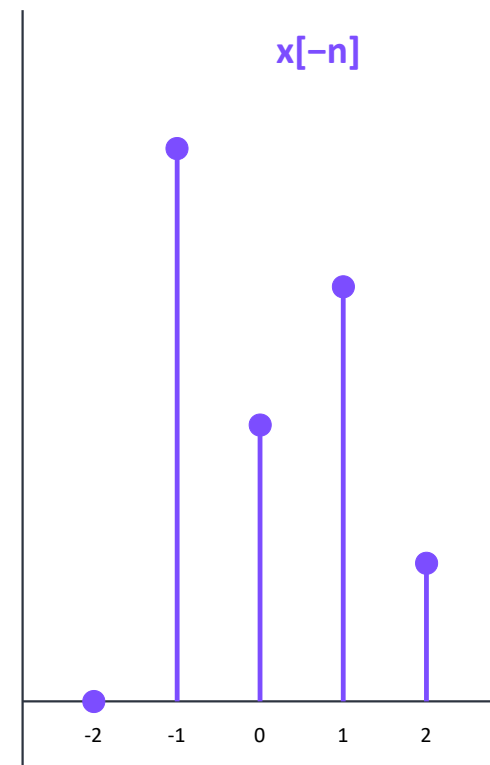
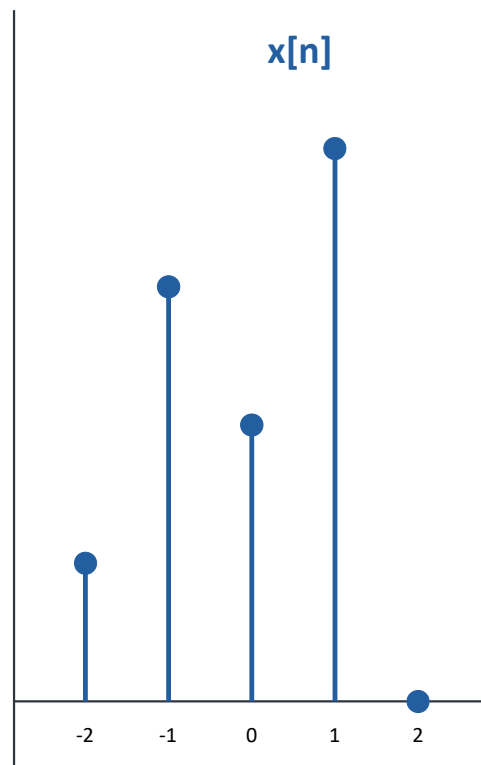
$$y[n] = x[-n]$$

Reflect the sequence about $n = 0$.

Sample at $n = +2$ moves to $n = -2$.

Sample at $n = -1$ moves to $n = +1$.

The order of amplitudes reverses around the origin.



Order Matters in Combined Time Operations

Example:
 $y[n] = x[-2n + 4]$

**Recommended
approach**

1. Rewrite: $-2n + 4 = -2(n-2)$

2. First apply folding + compression conceptually

3. Then apply the shift carefully

4. Verify by mapping a few individual indices

Classification of Discrete-Time Signals

Periodic/ aperiodic

Energy/Power

Deterministic / Random

Static / Dynamic

Time Variant/ invariant

Linear / Non-linear

Causal/non-causal

Stable/ Unstable

Periodic and Aperiodic Sequences

Periodic if:
 $x[n + N] = x[n]$
for some positive integer N

**The smallest positive N is the
fundamental period N_0 .**

Example: $\cos(\pi n/2)$ has period $N_0 = 4$.

Discrete-time sinusoid is periodic only if $\omega_0/(2\pi)$ is rational.

Otherwise, the sequence is aperiodic.

Energy of a Discrete-Time Signal

Energy E of a signal $x[n]$ is

$$E \equiv \sum_{n=-\infty}^{\infty} |x(n)|^2$$

- An energy signal has finite, nonzero total energy: $0 < E < \infty$.
- Applies for complex signals as well
- The squared magnitude measures contribution from each sample.
- For finite-energy, finite-duration signals, we normally talk about **energy**
- But many important signals — periodic signals, step functions, random noise, sinusoids — go on forever and never die out.
- If you try to sum $|x(n)|^2$ over all n from $-\infty$ to ∞ , the sum **diverges to infinity**. Energy becomes meaningless for these signals.

Average Power of a Discrete-Time Signal

- So instead of energy, we ask: *"On average, how much energy does this signal deliver per sample?"*
That's **power** — energy per unit time, averaged over an (eventually infinite) observation window.

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

- A power signal has finite, nonzero average power
- Periodic sequences are commonly power signals
- If we define signal energy of $x(n)$ over finite interval $-N \leq n \leq N$ as

$$E_N \equiv \sum_{n=-N}^N |x(n)|^2$$

then signal energy $E = \lim_{N \rightarrow \infty} E_N$

and power of a signal $x(n)$ is

$$P \equiv \lim_{N \rightarrow \infty} \frac{1}{2N+1} E_N$$

- If E is finite, $P=0$
- If E is infinite, avg. power may be finite or infinite

1. Finite energy → zero average power

- Suppose a signal exists only for a finite duration, such as a **camera flash, radar pulse, or a short voltage pulse**.
- It may contain a finite amount of energy: $0 < E < \infty$
- But if we observe it over an increasingly long time, the energy gets spread over a larger and larger observation interval, so $P=0$

Physical interpretation:

The signal delivers a finite amount of energy, but it does not continuously deliver energy.

For example, imagine a 10-Joule pulse lasting 1 second. The total energy is 10 J. But if you calculate its average power over 1000 seconds:

$$P_{\text{avg}} = 10/1000 = 0.01 \text{ W}$$

and as the observation time approaches infinity,

$$P_{\text{avg}} \rightarrow 0.$$

So, energy signals are short bursts of activity

2. Infinite energy → average power may be finite

- A sinusoidal signal: $x[n] = A \cos(\omega_0 n)$
- It continues forever, so it has infinite total energy: $E = \infty$
- But its average power is finite: $P = A^2/2$ for a sinusoid of amplitude A .

Physical interpretation:

A continuously operating AC voltage does not stop delivering energy. Hence its total energy over infinite time is infinite. However, the **rate at which it delivers energy** remains finite.

This is exactly analogous to an electrical appliance consuming **100 W continuously**:

After 1 second → 100 J

After 1 hour → 360,000 J

After infinite time → infinite energy

But its power remains: **100 W**

So, power signals are continuously operating systems.

Average Power of a Discrete-Time Signal

1. $|x(n)|^2$ — instantaneous energy/power at sample n .
 2. $\sum_{n=-N}^N |x(n)|^2$ — total energy in a finite window
 3. $\frac{1}{2N+1}$ — normalize by window length
 4. $\lim_{N \rightarrow \infty}$ — extend the window to cover the entire signal
1. Squared magnitude of the signal at each time index. Squaring makes it non-negative (energy/power is always ≥ 0) and works for complex-valued signals too
 2. This adds up the "energy" contained in a symmetric window of samples, from $n=-N$ to $n=N$. This window contains $2N+1$ samples total (count: N negative indices, N positive indices, plus the single $n=0$ sample)
 3. Dividing by the number of samples turns the total energy in the window into an **average energy per sample** — i.e., the average power *within that window*.
 4. As $N \rightarrow \infty$, the window expands to include the whole signal (past and future). Taking the limit gives you the **long-run average power** — a finite number, even though the total energy E may be infinite.